IB MATH AA SL

Lesson 41

Integration & Antiderivatives

Calculus · 60-minute lesson

Syllabus SL 5.5 ·

Do Now — recall

ON YOUR WHITEBOARD

1. Differentiate x^3.

2. So what differentiates to give $3x^2$?

3. What differentiates to give $2x$?

Answers (click to reveal)

1. $3x^2$
2. x^3
3. x^2

What we're learning today

LEARNING OBJECTIVES

1. **Understand**
integration as the reverse of differentiation.

2. **Use**
the power rule for integration.

3. **Include**
the constant of integration C.

4. **Integrate**
sums of power terms.

SUCCESS CRITERIA — I CAN...

✓ explain integration as 'anti-differentiation'

✓ use $\int x^n \, dx = x^{n+1}/(n+1) + C$

✓ always add + C for an indefinite integral

✓ integrate term by term

Reverse the power rule

THE RULE

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

Add 1
increase the power by 1

Divide
by the new power

+ C
always, for an indefinite integral

WHY + C

Differentiating a constant gives 0,

so the constant is 'lost'.

Restore it
with + C.

Integrate a polynomial

QUESTION

Find $\int (6x^2 + 2)\,dx$.

SOLUTION

Integrate each term (add 1, divide):

$$\int 6x^2\,dx = \frac{6x^3}{3} = 2x^3$$

$\int 2\,dx = 2x$. Don't forget + C:

$$= 2x^3 + 2x + C$$

Your turn — integrate

ANSWER IN YOUR BOOK

1. $\int x^3\,dx$.

2. $\int 4x\,dx$.

3. $\int (2x + 1)\,dx$.

Answers (click to reveal)

1. $x^4/4 + C$
2. $2x^2 + C$
3. $x^2 + x + C$

Finding C

USE A KNOWN POINT

$$\int 2x \, dx = x^2 + C$$

Integrate to get the general expression

Substitute a known point to find C

EXAMPLE

dy/dx = 2x

→ $y = x^2 + C$

passes (1, 5)

5 = 1 + C → C = 4

so

$y = x^2 + 4$

Find the constant

QUESTION

A curve has $dy/dx = 2x$ and passes through (3, 10).

Find y.

SOLUTION

Integrate: $y = x^2 + C$.

Use (3, 10): $10 = 3^2 + C \rightarrow C = 1$:

$$y = x^2 + 1$$

EXAM-STYLE QUESTION

Exam-style question

(a) Find $\int (3x^2 - 4x + 5)\,dx$. [3]

(b) The gradient of a curve is $dy/dx = 4x$ and it passes through $(1, 7)$. Find y. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $x^3 - 2x^2 + 5x + C$ (A1)(A1)(A1)

(b) $y = 2x^2 + C$ (M1); $7 = 2 + C \rightarrow C = 5$ (A1) $\rightarrow y = 2x^2 + 5$ (A1)

Common mistakes

× **+ C**
Never omit the constant for an indefinite integral.

× **Add then divide**
Increase the power by 1, then divide by the new power.

× **Integration ≠ differentiation**
They go in opposite directions.

× **Find C**
Use the given point to determine C when asked for y.

Mixed practice

QUESTIONS

1. $\int x^5\,dx$.

2. $\int 6x^2\,dx$.

3. $\int (x - 3)\,dx$.

4. $dy/dx = 3$, through $(0, 2)$. Find y.

Answers (click to reveal)

1. $x^6/6 + C$
2. $2x^3 + C$
3. $x^2/2 - 3x + C$
4. $y = 3x + 2$

Plenary & exit ticket

KEY FORMULAS

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$always \quad + C$$

EXIT TICKET — before you leave

1. $\int x^2 \, dx$.
2. $\int 2 \, dx$.
3. What must you always add?

Answers (click to reveal)

1. $x^3/3 + C$
2. $2x + C$
3. $+ C$

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Integration & antiderivatives.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 42

Definite integrals & the area under a curve.

$$\int_a^b f(x)\,dx$$